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Continuous Mapping Thm:

↳ convergence in Pr:

→ $Z_n \in \mathbb{R}^k$ $\quad g(u): \mathbb{R}^k \to \mathbb{R}^q$

if $Z_n \xrightarrow{p} c$ and $g(u)$ is cts at c,

$g(Z_n) \xrightarrow{p} g(c)$

↳ convergence in dist.:

→ if $Z_n \xrightarrow{d} Z$ and g has set of discontinuity points D_g s.t. $P(Z \in D_g) = 0 \Rightarrow g(Z_n) \xrightarrow{d} g(Z)$

Slutsky's:

↳ $Z_n \xrightarrow{d} Z$; $W_n \xrightarrow{p} c$

↳ $Z_n + W_n \xrightarrow{d} Z + c$ as $n \to \infty$

↳ $Z_n W_n \xrightarrow{d} Zc$ as $n \to \infty$

CMT:

$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \xi \Rightarrow g(\sqrt{n}(\hat{\theta} - \theta)) \xrightarrow{d} g(\xi)$

Delta Method:

↳ $\mu \in \mathbb{R}^k$; $g(u): \mathbb{R}^k \longrightarrow \mathbb{R}^q$

$$\sqrt{n}\,(g(\hat{\mu}) - g(\mu))$$

If $\sqrt{n}\,(\hat{\mu} - \mu) \xrightarrow{d} \xi$ and $g(u)$ is ctsly differentiable in a neighborhood of μ, then $\sqrt{n}\,(g(\hat{\mu}) - g(\mu)) \xrightarrow{d} \underline{G}^T \xi$ where $\underline{G}(u) = \frac{\partial g(u)^T}{\partial u}$ and $\underline{G} = \underline{G}(\mu)$

Stochastic Order Symbols:

↳ $Z_n = o_p(1)$; $Z_n = O_p(1)$

$Z_n \xrightarrow{p} 0$; bounded in probability (behaves like a number)

↳ $O_p(1) + o_p(1) = O_p(1)$

↳ $O_p(1) \cdot o_p(1) = o_p(1)$

Assumptions:

1. $\{(Y_i, X_i)\}_{i=1}^n$ are i.i.d
2. $E[Y^2] < \infty$
3. $E\|X^2\| < \infty$
4. $E[XX^T]$ is PD

$$\hat{\beta} = (\underline{X}^T\underline{X})^{-1}\underline{X}^T\underline{Y}$$
$$= \left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i^T\right)^{-1} \frac{1}{n}\sum_{i=1}^{n} X_i Y_i$$

Consistency of $\hat{\beta}$:

$\hookrightarrow$ Step 1: LLN: $\frac{1}{n}\sum_{i=1}^{n} X_i X_i^T \xrightarrow{p} E[XX^T]$

$$\frac{1}{n}\sum_{i=1}^{n} X_i Y_i \xrightarrow{p} E[XY]$$

$\hookrightarrow$ Step 2: CMT: $\hat{\beta} = g\left(\frac{1}{n}\sum_{i=1}^{n} (X_i X_i^T)^{-1}, \frac{1}{n}\sum_{i=1}^{n} X_i Y_i\right)$

where $g(\underline{A}, b) = \underline{A}^{-1} b$

$$\Rightarrow \hat{\beta} \xrightarrow{p} g(E[XX^T], E[XY]) = E[XX^T]^{-1} E[XY] = \beta$$

Assumptions:

① $\{(Y_i, X_i)\}_{i=1}^{n}$ are i.i.d

② $E[Y^4] < \infty$

③ $E\|X\|^4 < \infty$

} second moment of second moment

④ $E[XX^T]$ is PD

Asymptotic Normality: $\hat{\beta} = (\underline{X}^T\underline{X})^{-1}\underline{X}^T\underline{Y}$; $Y_i = X_i^T\beta + e_i$

$$= \left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i^T\right)^{-1} \frac{1}{n}\sum_{i=1}^{n} X_i Y_i$$

$$\hat{\beta} = \left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i^T\right)^{-1} \frac{1}{n}\sum_{i=1}^{n} X_i (X_i^T\beta + e_i)$$
$$= \beta + \left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i^T\right)^{-1} \underbrace{\frac{1}{n}\sum_{i=1}^{n} X_i e_i}_{op(1)}$$

$$\sqrt{n}(\hat{\beta}-\beta) = \left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i^T\right)^{-1} \frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i e_i$$

↳ Step I : CLT (a)

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i e_i = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i e_i\right) - \underbrace{E[Xe]} \Big) \longrightarrow N(0, \underline{\Omega})$$

$$\underset{k \times k}{\underline{\Omega}} = var(Xe) = E[XX^T e^2] \overset{\sim}{} 0$$

(b) LLN: $\frac{1}{n}\sum_{i=1}^{n} X_i X_i^T \xrightarrow{p} E[X_i X_i^T]$

↳ Step 2: CMT:

$$\sqrt{n}(\hat{\beta}-\beta) \xrightarrow{d} E[XX^T]^{-1} N(0, \underline{\Omega}) = N(0, \underline{V}_\beta)$$

where $\underline{V}_\beta = E[XX^T]^{-1} \underline{\Omega}\, E[XX^T]^{-1}$

→ $\underline{V}_\beta$ is asymptotic variance

↳ sandwich form

Homoskedasticity:

$$↳ E[e^2 | X] = \sigma^2 \Rightarrow \underline{\Omega}_0 = E[XX^T e^2]$$

$$= E[XX^T E[e^2 | X]]$$

$$= \sigma^2 E[XX^T]$$

$$\underline{V}_0 = \sigma^2 E[XX^T]^{-1} E[XX^T] E[XX^T]^{-1} = \sigma^2 E[XX^T]^{-1}$$

Estimating Asymptotic Variance under Homoskedast.

$\hookrightarrow$ estimate : $E[XX^T]$ by $\frac{1}{n}\sum_{i=1}^{n} X_i X_i^T$

$\sigma^2 : E[e^2]$ by $\frac{1}{n}\sum_{i=1}^{n} e_i^2 \xrightarrow{p} \sigma^2$

feasible: $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} \hat{e}_i^2$ s.t. $\hat{e}_i = Y_i - X_i^T\hat{\beta}$

$= X_i^T\beta + e_i - X_i^T\hat{\beta}$

$= e_i - X_i^T(\hat{\beta} - \beta)$

$$\hat{e}_i^2 = e_i^2 - 2e_iX_i^T(\hat{\beta}-\beta) + (\hat{\beta}-\beta)^T X_iX_i^T(\hat{\beta}-\beta)$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\hat{e}_i^2 = \frac{1}{n}\sum_{i=1}^{n} e_i^2 - 2\left(\frac{1}{n}\sum_{i=1}^{n} e_iX_i^T\right)\underbrace{(\hat{\beta}-\beta)}_{\to 0} + \underbrace{\underbrace{(\hat{\beta}-\beta)^T}_{\to 0}\underbrace{\left(\frac{1}{n}\sum_{i=1}^{n} X_iX_i^T\right)}_{E[XX^T]}\underbrace{(\hat{\beta}-\beta)}_{\to 0}}_{op(1)}$$

$$\frac{1}{n}\sum_{i=1}^{n}\hat{e}_i^2 \xrightarrow{p} \sigma^2 + op(1) = \hat{\sigma}^2$$

or

$$\hat{\sigma}^2 \xrightarrow{p} \sigma^2$$